Hyperparameter Comparison for Reproduction Audit

Inventory uses the paper decision-quality target. Cliff uses Appendix Table 3, not the separate Cliff Figure 2 decision-quality panel. All listed Cliff methods are trained from scratch by the reproduction runner.

Item	paper-inventory	repro-inventory	paper-cliff	repro-cliff
State space size	36, $s_{\max} = 30$, $b_{\max} = 5$	same	48, 4×12 grid	same
Augmented Lagrangian ρ for DFL-LP	10	10^4	0.1	0.1
Augmented Lagrangian ρ for DFL-Sketch	0.1	10	1.0	0.1
Augmented Lagrangian ρ for DFL-QP	10^4	10^4	0.1	0.1
QP regularization η / `qp_reg`	10^{-3}	**50**	0.1	0.1
Sketch ensemble size K	8	8	8	**4**
Sketch row-keep ratio α	0.1	0.1 for Sketch; 0.2 for SA configs	0.2	**0.1**
LP basis ridge	10^{-6}	not explicitly changed in repro runner	none	0.0 for SA
Aggregation sizes	$M \in \{10, 20, 30\}$	same	Table 2 lists $M \in \{10, 20, 30\}$; Appendix Table 3 target uses $M \in \{6, 12, 24\}$	$M \in \{6, 12, 24\}$
Softmax temperature τ	0.5	default / not separately overridden in the inventory runner	0.5	$M = 6$: 0.5; $M = 12$: 0.5; $M = 24$: 2.0
SA loss weight	not clearly listed	1	not clearly listed	**500**
Train samples	10	10	100	100
Test samples	50	50	100	100

Parameters that need confirmation.

- Inventory `qp_reg`: Table 2 gives 10^{-3}, while the local reproduction recipe uses 50.
- Inventory Sketch ρ: Table 2 gives 0.1, while the local reproduction recipe uses 10.
- Cliff Sketch settings: Table 2 gives $K = 8$ and row-keep ratio 0.2, while the from-scratch Table 3 recipe uses $K = 4$ and row-keep ratio 0.1.
- Cliff SA loss weight: the local Appendix Table 3 reproduction uses 500, but this value is not clearly listed in Table 2.
- Cliff SA temperature: Table 2 gives $\tau = 0.5$, while the local Table 3 recipe uses $\tau = 2.0$ for $M = 24$.
- Cliff aggregation sizes: Table 2 lists $\{10, 20, 30\}$, while Appendix Table 3 uses $\{6, 12, 24\}$.